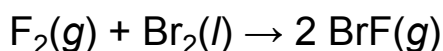


Elemental fluorine and bromine react to form the diatomic interhalogen compound bromine monofluoride according to the equation below.



Which of the following statements correctly describes the reaction?

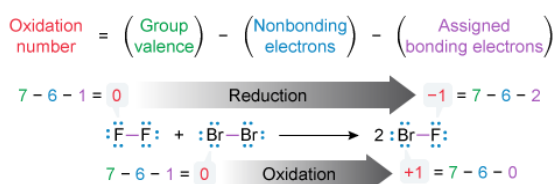
- ☐ A. The fluorine atoms are oxidized, and their oxidation number changes from 0 to +1. [6%]
- ☐ B. The fluorine atoms are reduced, and their oxidation number changes from 0 to -2. [11%]
- ☒ C. The bromine atoms are oxidized, and their oxidation number changes from 0 to +1. [71%]
- ☐ D. The bromine atoms are reduced, and their oxidation number changes from 0 to -1. [10%]

Correct

72% answered correctly

Explanation:

Changes in oxidation numbers indicate oxidation or reduction



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Oxidation-reduction (redox) changes occurring within a reaction can be determined by evaluating the oxidation number of each atom in the product and reactant species. The **oxidation number** (oxidation state) of an atom accounts for any gain or loss of electrons by the atom relative to the number of valence electrons present in its elemental state (ie, its **group valence number**). The oxidation numbers can be determined using **oxidation state rules** or calculated according to the equation:

$$\text{Oxidation number} = \left(\text{Group valence} \right) - \left(\text{Nonbonding electrons} \right) - \left(\text{Assigned bonding electrons} \right)$$

When calculating an oxidation number, all bonding electrons (two electrons

per bond line) are assigned to the most **electronegative** atom in a bond. This is done because the oxidation number calculation treats all bonds as if they are fully ionic. The bonding electrons are evenly divided between the atoms only if the electronegativities of the bonded atoms are equal.

It is important to recognize that the oxidation number of an atom is conceptually different from its **formal charge**, which indicates charge distribution among atoms in a structure. The formal charge calculation always divides the bonding electrons in half because it treats all bonds as if they are fully covalent (sharing electrons equally). As such, the oxidation number and formal charge of an atom are not necessarily the same value.

By **comparing the oxidation number** for a given element in the reactants with the oxidation number of that same element in the products, a loss of electrons (oxidation) or a gain of electrons (reduction) can be identified. Elements that have oxidation numbers that do not change are not involved in redox processes.

In the given reaction, the bromine atoms in BrF are assigned 0 bonding electrons because fluorine is more electronegative than bromine. Therefore, the bromine atoms are oxidized and go from their elemental state in Br₂(l) (ie, an **oxidation number of 0**) to an oxidation number of +1 in BrF(g).

$$\begin{array}{l} \text{Oxidation number} \\ (\text{Br in Br}_2) \end{array} = (7) \begin{array}{l} \text{Group} \\ \text{valence} \end{array} - (6) \begin{array}{l} \text{Nonbonding} \\ \text{electrons} \end{array} - (1) \begin{array}{l} \text{Assigned} \\ \text{bonding electrons} \end{array} = 0$$

$$\begin{array}{l} \text{Oxidation number} \\ (\text{Br in BrF}) \end{array} = (7) \begin{array}{l} \text{Group} \\ \text{valence} \end{array} - (6) \begin{array}{l} \text{Nonbonding} \\ \text{electrons} \end{array} - (0) \begin{array}{l} \text{Assigned} \\ \text{bonding electrons} \end{array} = +1$$

(Choice A) The fluorine atoms are *reduced* in the reaction.

(Choice B) Although the fluorine atoms are reduced in the reaction, their oxidation number changes from 0 to -1 (not -2).

(Choice D) The bromine atoms are *oxidized* in the reaction.

Educational objective:

By comparing the oxidation number for a given element in the reactants with the oxidation number of that same element in the products, oxidation and reduction processes can be identified.

Time spent:0 sec

Subject:

General Chemistry

QID:403893

System:

Interactions of Chemical Substances